Proposition ①

Suppose that $\{v_1, v_2, \dots, v_m\} \subseteq V$ is a set of non-zero eigenvectors for T with distinct eigenvalues

i.e. • $T(v_i) = \lambda_i v_i$, $\lambda_i \in F$
• $\lambda_i \neq \lambda_j$ when $i \neq j$

Then $\{v_1, v_2, \dots v_m\}$ is a linearly independent set

Pf: If they were not linearly independent, we could (after reindexing) write

$$\lambda_1 \cdot v_1 = \lambda_1 [a_2 v_2 + a_3 v_3 + \dots + a_m v_m]$$

one of these is non-zero

$T(\cdot)$

$$T(v_1) = a_2 T(v_2) + a_3 T(v_3) + \dots + a_m T(v_m)$$

$$\| \qquad\qquad \|$$

$$\lambda_1 v_1 = a_2 \lambda_2 v_2 + a_3 \lambda_3 v_3 + \dots + a_m \lambda_m v_m$$

$$\| \qquad\qquad //$$

$$a_2 \lambda_1 v_2 + a_3 \lambda_1 v_3 + \dots + a_m \lambda_1 v_m$$

$$a_2 \overbrace{(\lambda_1 - \lambda_2)}^{\neq 0} v_2 + a_3 \overbrace{(\lambda_1 - \lambda_3)}^{\neq 0} v_3 + \dots + a_m \overbrace{(\lambda_1 - \lambda_m)}^{\neq 0} v_m = 0$$

one of these is non-zero

$\Rightarrow$ $\{v_2, \dots, v_m\}$ are linearly dependent

Repeating the argument, and reindexing again, we find $\{v_3, \dots, v_m\}$ are linearly dependent

In fact, we could have presented this proof as a proof by induction

$P(m)$: Any set $\{v_1, \ldots, v_m\}$ of eigenvectors for V with distinct eigenvalues is linearly independent.

Base case: $(m = 1)$ ✓ eigenvectors are non-zero

Inductive hypothesis: $P(m-1)$ is true

Inductive step: $P(m)$ is not true

$\Longrightarrow P(m-1)$ is not true

contra-diction

Corollary (1): $T: V \to V$

If $\dim V = n$ & we have

$(v_1, \dots, v_n)$: eigenvectors with distinct eigenvalues

Then $(v_1, \dots, v_n)$ is an eigenbasis for T.

Corollary (2): $A \in M_n(F)$

$ch_A(x)$ has n distinct zeros in F. Then A has an eigenbasis $\iff$ A is diagonalizable

Rmk: This is not a necessary condition for diagonalizability

I_n only has one eigenvalue

but is diagonalizable.

Example: ① $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ also has one e. value

and is not diagonalizable.

② $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \in M_{2\times 2}(\mathbb{R})$

is not diagonalizable because

$ch_A(x) = x^2 + 1$ has no zeros in $\mathbb{R}$

However in $\mathbb{C}$, $ch_A(x) = (x+i)(x-i)$

$\Rightarrow A \in M_{2\times 2}(\mathbb{C})$ has an eigenbasis in $\mathbb{C}^2$.

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \underbrace{\begin{bmatrix} i \\ 1 \end{bmatrix}}_{v_1} = \begin{bmatrix} -1 \\ i \end{bmatrix}$$

$$= i \begin{bmatrix} i \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \underbrace{\begin{bmatrix} -i \\ 1 \end{bmatrix}}_{v_2} = -i \begin{bmatrix} -i \\ 1 \end{bmatrix}$$

$$A = \underbrace{\begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix}}_{C} \underbrace{\begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}}_{D} \underbrace{\begin{bmatrix} -\frac{i}{2} & \frac{1}{2} \\ \frac{i}{2} & \frac{1}{2} \end{bmatrix}}_{C^{-1}}$$

$Ce_1 = v_1$ $\qquad$ $C^{-1}v_1 = e_1$

$Ce_2 = v_2$ $\qquad$ $C^{-1}v_2 = e_2$

Thm (Fundamental thm of algebra)

Every non-constant polynomial $p(x) \in Poly(\mathbb{C})$ is a product of linear polynomials.

Corollary (3) $A \in M_{n \times n}(\mathbb{C})$

$$ch_A(x) = (-1)^n \prod_{i=1}^{r} (x - \lambda_i)^{m(\lambda_i)}$$

where $\lambda_1, \ldots, \lambda_r \in \mathbb{C}$ are distinct

$m(\lambda_i) \in \mathbb{N}$

Notation: $\prod_{i=1}^{r} c_i = c_1 c_2 \cdots c_r$

Example: $ch_{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}}(x) = (x-1)^2$

$\lambda_1 = 1, \quad m(\lambda_1) = 2.$

Example:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad ch_A(x) = (-1)^3 (x-1)^3$$

$$B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad ch_B(x) = (-1)^3 (x-1)^3$$

$$C = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad ch_C(x) = (-1)^3 (x-1)^3$$

$\ker(B - I_3)$

$$\ker \begin{array}{c} \begin{array}{ccc} a & b & c \end{array} \\ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{array} = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} : b = 0 \right\}$$

$$= \left\{ \begin{bmatrix} a \\ 0 \\ c \end{bmatrix} = a e_1 + c e_3 \quad a, c \in \mathbb{C}^2 \right\}$$

NOT DIAGONALIZABLE

$$\ker(K - I_3)$$

$$\ker \begin{array}{c} \begin{array}{ccc} a & b & c \end{array} \\ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{array} = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} : \begin{array}{l} b = 0 \\ c = 0 \end{array} \right\}$$

$$= \left\{ \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix} = a \cdot e_1 \quad a \in \mathbb{C} \right\}$$